



Design And Test Report by

YUVAAN

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00 Introduction



0.1 Introduction

This System Design and Development Review (SDDR) Report documents the comprehensive process and methodologies employed by our team in the design and development of "Yuvaan", a rover specifically made according to the problem statement given for the International Rover Challenge (IRC) by Space Robotics Society (SPROS).

Our team's primary objective was to design a versatile rover equipped to navigate and explore uncharted terrains. It features capabilities for object retrieval, transport, and maintenance tasks through a robotic arm, along with soil sample collection and analysis via an integrated astrobiology testing module.



0.2 Project Management

Position	Responsibility
Mentor / Faculty Advisor	Coordinates access to institutional resources, including 3D printing and Central Workshop machinery. Provides technical guidance and expert advice to support the student team in overcoming challenges.
Team Lead	Oversees and coordinates subteams to ensure a seamless, modular workflow across the project. Manages essential non-technical aspects, including material procurement, arrangement of manufacturing facilities and more
Heads	Designated module leads who support the team lead in rover development, facilitating effective coordination with sub-teams. Each lead oversees specific modules, ensuring integration and alignment with project goals.

Module / Sub Team	Responsibility
Chassis and Mobility	Point of contact for design, construction, and testing of the rover's structural and mobility components, ensuring robust and adaptable movement across terrains.
Manipulator System	Responsible for the development and integration of the robotic arm, including object retrieval, handling tasks, and coordination with other subsystems.
Astrobiology Module	Leads the design and functionality of the onboard soil sampling and analysis unit, collaborating on sample testing and data gathering.
Electronics and Power	Oversees all electrical systems, power distribution, and energy management, ensuring reliable and efficient operation.
Communication Systems	Manages wireless protocols, signal integrity, and communication reliability for remote control and data transmission.
Software and Control	Develops control algorithms, software architecture, and system integration, enabling responsive and autonomous rover functions.



0.3 Project Planning

Phase	Timeline	Description
Project Decision	July 2025 (1st week)	Decided to participate in SPROS IRC a
Team Foundation	July 2025	Held meetings with team lead, heads, i recruited members for IRC 2025.
Recruitment	July - August 2025	Task-based submissions and personal i during summer break.
Initial Learning	August 2025	No team division; members explored i previous IRC submissions.
Research and Design	October 2025	Conducted research and design in CAD planning, and software design.
Manufacturing Start	October 2025	Began manufacturing key components
Current Phase	November 2025	Manufacturing and assembly phase; hi printing and finishing work.
Testing and Rectification	December 2025	Quality checks, mission testing, and re issues.
Project Conclusion	December 2025	Final assembly and testing. Ongoing s



0.4 Resource Management

Materials and Components

Name	Procurement Source
Rhino RMCS series motors and Cytron Motor Drivers	Robokits India
Gearbox FW-40-30-71B5	Local Market - Guwahati
Stainless Steel Sheet and pipes (rectangular and round), Aluminium Sheet, Mild Steel Rods, Nuts and Bolts	Local Market - Guwahati
Aluminium T slot extrusion rods and L - Joints	Robokits India
Batteries (6s)	Robokits India
3D printer filaments (TPU, PLA)	Local Market - Guwahati

Budget Allocation

Name	Estimated Cost (in Rupees)
Raw Materials	14,000
Machine Parts, Nuts and Bolts etc.	11,000
Motors, Microcontrollers, Jetson, Battery	1,30,000
Outsourcing	25,000
Total	1,80,000

01 MECHANICAL DESIGN

The mechanical design serves as the rover's engineered backbone, blending structural endurance, precision motion, and mission-grade adaptability into a unified physical platform. Its chassis, linkages, and suspension assemblies are optimized for maximum strength-to-weight efficiency, enabling controlled mobility across unpredictable terrains with minimal energy loss. Precision-machined joints and high-accuracy actuator interfaces empower the manipulator system to execute both force-intensive operations and delicate, fine-resolution tasks with confidence. Environmental sealing, modular mounting standards, and field-serviceable construction ensure long-term durability, rapid subsystem integration, and resilience against mechanical wear and harsh operating conditions. Together, this design elevates traditional mechanical engineering into a responsive, intelligent, and mission-ready system capable of thriving in demanding planetary and terrestrial environments.

1.1 CHASSIS

- **Core Function:** Serves as the primary load-bearing platform and core structural framework for all mechanical and electronic systems.
- **Construction:** Built with high-strength aluminium T-slot extrusions, offering a lightweight, modular design with dimensional stability.
- **Base Plate:** Aluminium sheet with dedicated slots for the accurate alignment and secure installation of the Astro-Bio drilling assembly.
- **Suspension Robustness:** Incorporates a Dual-Bearing Support System to handle uneven terrain loads.
- **Bearing Details:** Uses dual self-aligning pillow-block bearings mounted on a reinforced T-slot cross-member to provide stable two-point support for the suspension shaft.
- **Performance Benefit:** This minimizes shaft deflection, reduces vibration, and ensures smoother, more reliable suspension movement for enhanced long-term durability.

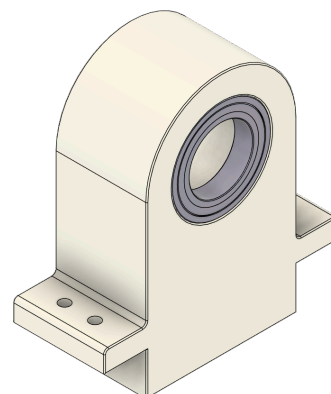
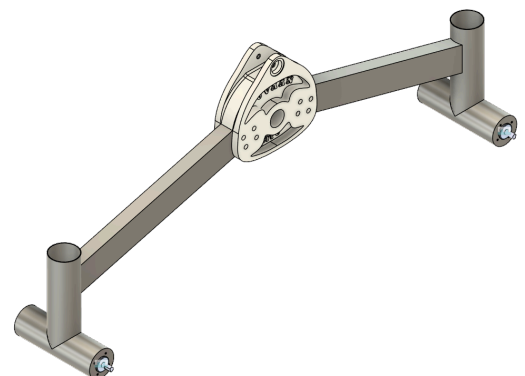
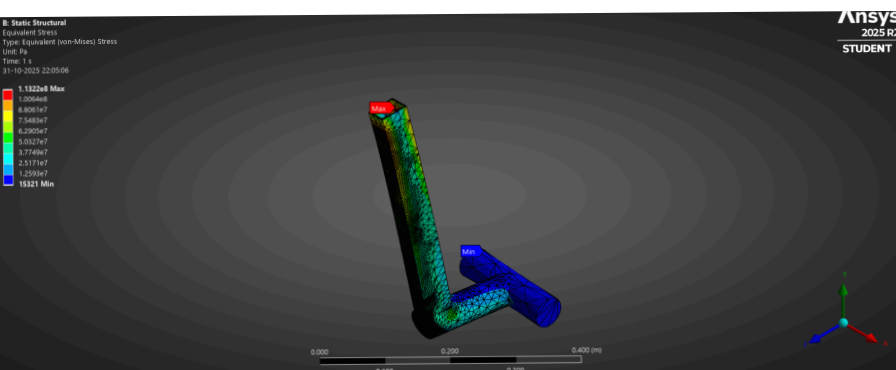
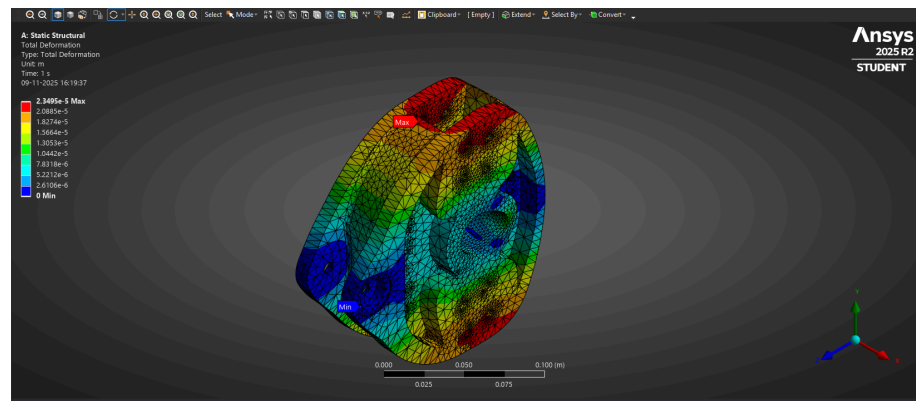
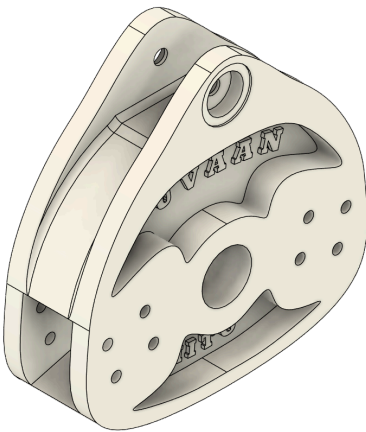


Figure 1.1

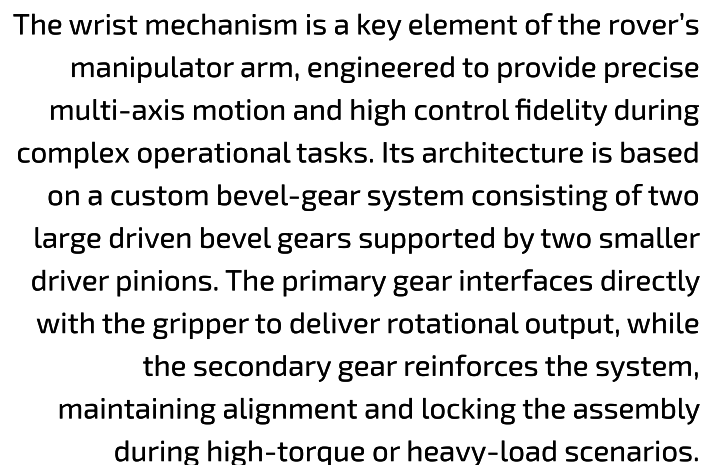
1.2 SUSPENSION

- The rover utilizes a Rocker-Bogie articulated suspension system with two independent rigid angular linkages capable of rotational displacement about a lateral axis parallel to the chassis base.
- This mechanism maintains continuous wheel-terrain contact, ensuring high traction across uneven surfaces while preserving ground clearance and uniform load distribution.
- Each suspension linkage is designed to withstand a nominal load of 32.5 kg (50% of total rover mass), with a factor of safety of 2.27, resulting in a final design load of ≈ 73.8 kg or ≈ 724 N under standard gravity.
- The suspension components are manufactured from SS-304 stainless steel, selected for its high tensile strength, abrasion resistance, and excellent corrosion durability in harsh environments.
- Using trigonometric optimization of angles θ (theta) and ϕ (phi), the geometry minimizes downward load transfer to the wheel subsystem, reducing excessive normal forces and preventing subsidence on loose or granular terrain.
- The optimized angular configuration maximizes ground clearance, maintains stable axle spacing throughout travel, and ensures smooth, controlled suspension articulation during traversal.





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- A 3D perspective view of a mechanical linkage mechanism. The mechanism consists of several interconnected bars and joints. Key dimensions are indicated: a 115° angle between two bars, a 120° angle between two other bars, and a 120° angle between two more bars. A dimension of 120 is also shown. The mechanism is labeled with 'v2.1' and 'v2.2'.

DRILL MECHANISM AND SOIL COLLECTION

Our soil extraction system employs an auger drill bit and a linear actuator for efficient collection. The process includes:

- **Drilling:** A linear actuator drives the drill vertically to penetrate the soil.
- **Lifting:** Drill retracts without rotation, retaining soil on the bit.
- **Discharge:** Drill rotates upon retraction, releasing soil tangentially.
- **Collection:** Discharged soil is funnelled into designated containers.

This method ensures efficient soil retrieval and controlled sample collection.

Testing Mechanism

The testing mechanism consists of a plate carrying six beakers, and a syringe mechanism for the flow of chemicals. The chemicals from the syringe assembly will be dropped into four containers. After the complete reactions, the sample changes its colour. The spectrometer will detect the resulting colour, and the USB microscope will detect the texture of the soil.

Analysis on Sample

Detection of life chances is determined by the presence of some major organic compounds such as proteins, lipids and nitro compounds which are present in most living organisms.

Some tests are as follows:

- **Biuret Reagent Test:**
Biuret Reagent is used to test proteins that give Purple Colour a positive result.
- **Diphenylamine Nitrate Test :**
This test is used to test nitrates which gives Blue Colour as a positive result.
- **Sudan IV Stain Test :**
This test detects lipids which gives Reddish Brown Colour as a positive result.
- **Flavin adenine dinucleotide (FAD) is detected by fluorescence at 450 nm,**
which can be observed using a USB microscope and spectrometer.

These tests combined help to determine the presence of life or the chances of survival of living organisms by detecting the organic compounds mentioned above.

Sensors:

- pH Sensor
- USB Microscope
- BME 280 Sensor
- Soil Moisture Sensor
- Spectrometer
- NPK Sensor
- MQ2 A Gas Sensor

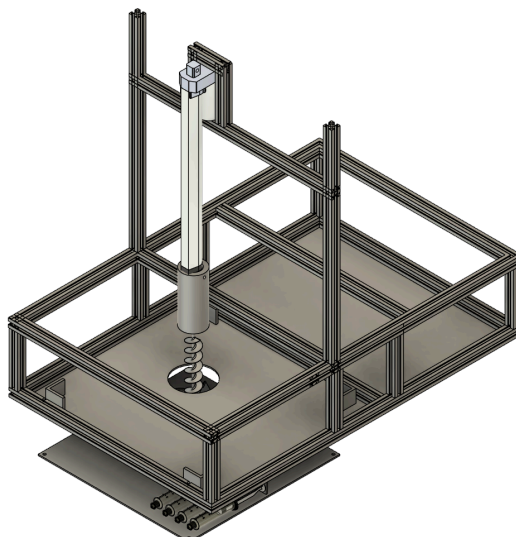


Figure 6.1

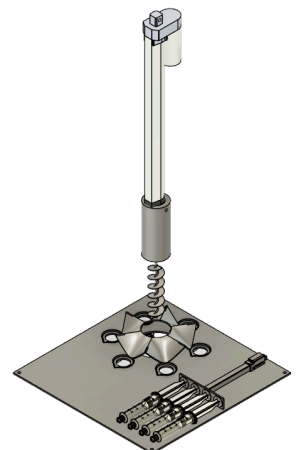


Figure 6.2

CONTROL FRAMEWORK

Our ROS Noetic-based control framework enables seamless communication between the Jetson module and workstation, supporting both Manual and Autonomous modes for mission flexibility.

Dual Operation Modes

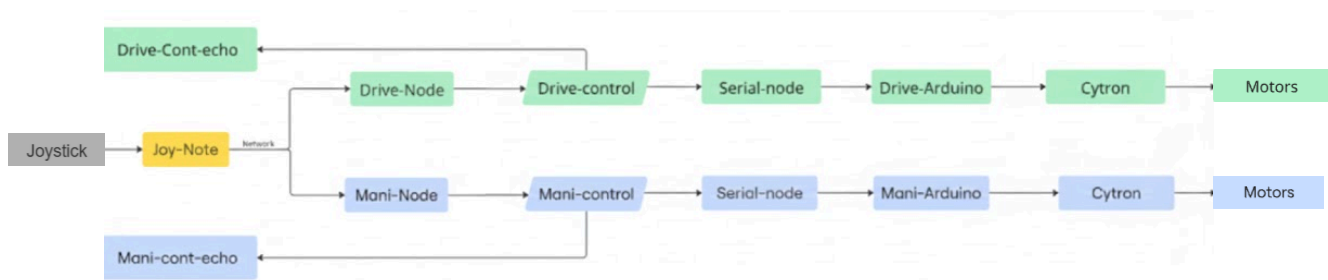
Autonomous Mode	Sensor-driven decision-making for independent navigation.
Manual Mode	Direct operator control via a gamepad interface.

Communication and Data Flow

ROS Noetic facilitates smooth data and command flow between the Jetson and workstation. This ensures efficient communication between input sources, processing nodes, and output components.

Data Flow Breakdown:

- Input Sources:
 - Gamepad
 - ZED Camera (and other sensors)
- Processing Nodes:
 - Data processing algorithms (for both manual and autonomous control)
- Output:
 - Motor control
 - Actuator control



Modular Framework

The system's modular nature allows the easy integration of new sensors or algorithms, enhancing its adaptability and minimising disruptions during updates or enhancements.

Modular Components:

- New sensors (e.g., LIDAR, IMU, GPS)
- New algorithms (e.g., path planning, sensor fusion)

This framework ensures that both operational modes work together effectively, fully aligned with mission objectives, and adaptable to changing requirements.

AUTONOMOUS ROVER NAVIGATION

The rover's navigation pipeline is based on a custom-developed autonomous navigation algorithm, enabling real-time path planning, obstacle avoidance, and pose correction. The framework integrates perception, localization, and control into a unified ROS-based system.

Perception and Obstacle Avoidance

- A Convolutional Neural Network (CNN) detects large boulders, rocks, and other obstructions in the rover's path.
- Depth and spatial context are obtained from camera and sensor data, enabling obstacle size estimation and avoidance feasibility.
- A real-time environmental map is generated, where detected obstacles dynamically update the local traversal cost.

The rover selects the next safe waypoint using a local decision layer that evaluates:

- Obstacle proximity
- Surface traversability
- Required deviation from the planned route

This ensures continuous, adaptive path correction without pre-mapped terrain.

Sensor Fusion and Localization

Accurate localization is achieved using an Extended Kalman Filter (EKF) fusion model implemented through the `robot_localization` package.

Sensors Used in Fusion:

Sensor	Data Provided	Purpose
IMU	Orientation & Angular Velocity	Stabilization and short-term orientation tracking
GPS	Global Position Fix	Long-range position reliability and drift correction
Wheel Odometry (Encoders)	Linear & Angular Displacement	Continuous motion estimation between GPS updates
Visual Data (Camera / Depth)	Feature-based motion cues	Enhances reliability in low-GPS / uneven terrain

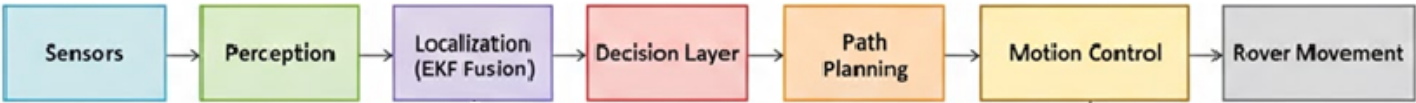
The EKF refines rover pose by combining high-frequency encoder + IMU data with slower, drift-correcting GPS and visual observations.

This ensures **precise and drift-minimized localization**, even in rugged terrain.

Path Planning and Motion Control

- The global mission route is broken into intermediate **target waypoints**.
- At each step, the custom navigation layer selects a **locally optimal movement vector** taking into account obstacle clearance and rover kinematics.
- Steering and velocity control commands are computed and transmitted to the motor drivers via the Jetson–Arduino control channel.

This custom algorithm enables **smooth, safe, and adaptive traversal**, without relying on predefined without relying on predefined.



MANUAL CONTROL SOFTWARE

Manual control enables the rover to be directly operated through a gamepad, with command signals processed on the workstation and transmitted to the Jetson for execution.

Control Flow

- 1. Gamepad Input:** Operator inputs are captured through a gamepad interface connected to the workstation.
- 2. Workstation Processing:** The workstation processes these inputs through the ROS Noetic framework, converting them into actionable commands.
- 3. Network Communication:** Commands are then transmitted over the network to the ROS framework on the Jetson module.
- 4. Serial Communication with Arduino:** The Jetson communicates with the Arduino controllers via serial communication, ensuring precise control of all connected components.
- 5. Actuation:** Commands are executed by the Arduino, directing the motors and components to perform as required based on the gamepad inputs.

Key Components of the Rover Control Centre

Component	Discription
Video Displayer	Displays a real-time panoramic video feed from the rover's cameras.
Command Buttons	Controls to manage rover mode and joystick command listening.
Rover Position and Speed	Displays real-time values of linear velocity, angular speed, and joint positions.
Output Terminal	Shows error messages and debugging information

Video Displayer

The Video Displayer fetches the video feed from the cameras mounted on the rover. Through the use of video stitching, it combines multiple camera streams to create a panoramic view. This video is then streamed in realtime on the website, providing live visual feedback of the rover's surroundings.

Command Buttons

There are two primary command buttons:

Start Button: When clicked, the server begins listening to joystick commands, allowing manual control of the rover.

Mode Button: This button allows the user to toggle between operational modes. The mode can also be changed directly from the joystick interface. The Mode Button enables the user to switch between manual control and autonomous mode seamlessly.

Rover Position and Speed

When the server listens to the joystick inputs, the website dynamically updates the following information:

Linear Velocity: Displays the current forward/backward speed of the rover.

Angular Speed: Shows the rover's turning speed (rotation rate).

Joint Positions: Displays the positions of the manipulator joints and the biological survey part of the rover.

Output Terminal

The Output Terminal is a console window that provides immediate feedback on the system's health. It displays:

Connection Errors: If there are issues with joystick connection or communication with ROS.

Programming Errors: Any issues or bugs in the ROS system will be shown here for quick debugging and troubleshooting

This feature ensures that the user can monitor the system status and take corrective actions without delay.

DIAGNOSTICS

The rover is equipped with an integrated diagnostics and monitoring framework designed to ensure reliable, safe, and efficient operation in dynamic environments. This system continuously observes the rover's mechanical, electrical, and computational subsystems through a network of sensors and analytical routines.

By combining real-time data acquisition, automated safety mechanisms, autonomous corrective behavior, and an operator-facing dashboard, the diagnostics architecture enables early fault detection, improves decision making, and enhances overall mission performance. It serves as the central backbone for maintaining system health, preventing failures, and supporting both autonomous and remote-controlled operation.

03 Electronics and Power System

The electronics and power system is designed to deliver reliable energy distribution and intelligent control across all rover sub-systems. The system ensures precise actuation for mobility, manipulation, and bio-assembly tasks. Integrated communication modules and emergency-protection features enhance operational safety and autonomy. Overall, the design enables efficient power management, seamless subsystem coordination, and robust performance in demanding mission environments.

5.1 Interdependencies Schematic

The diagram below shows a high-level overall System Architecture .

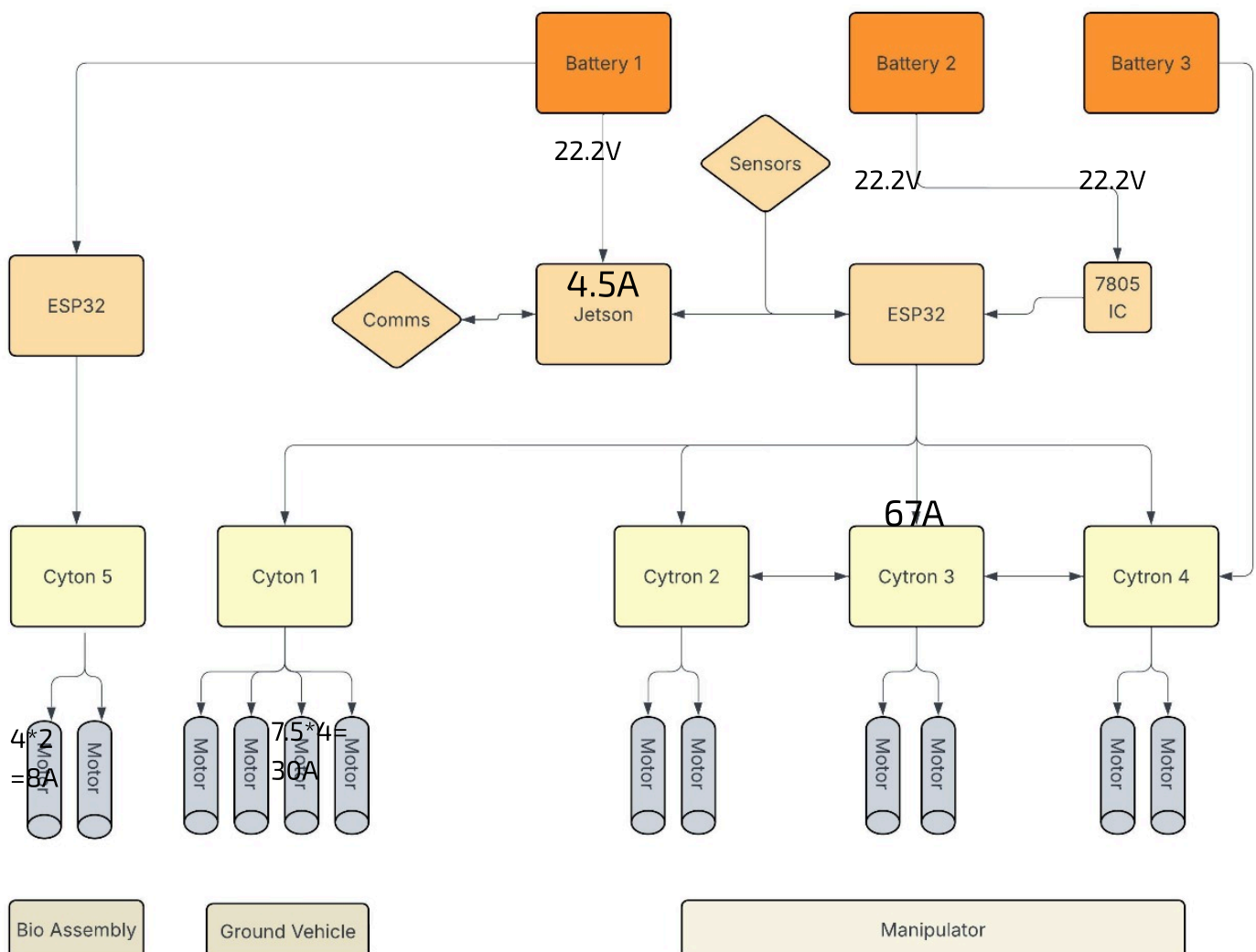


Figure 1.1

5.2 Hardware Description Schematic

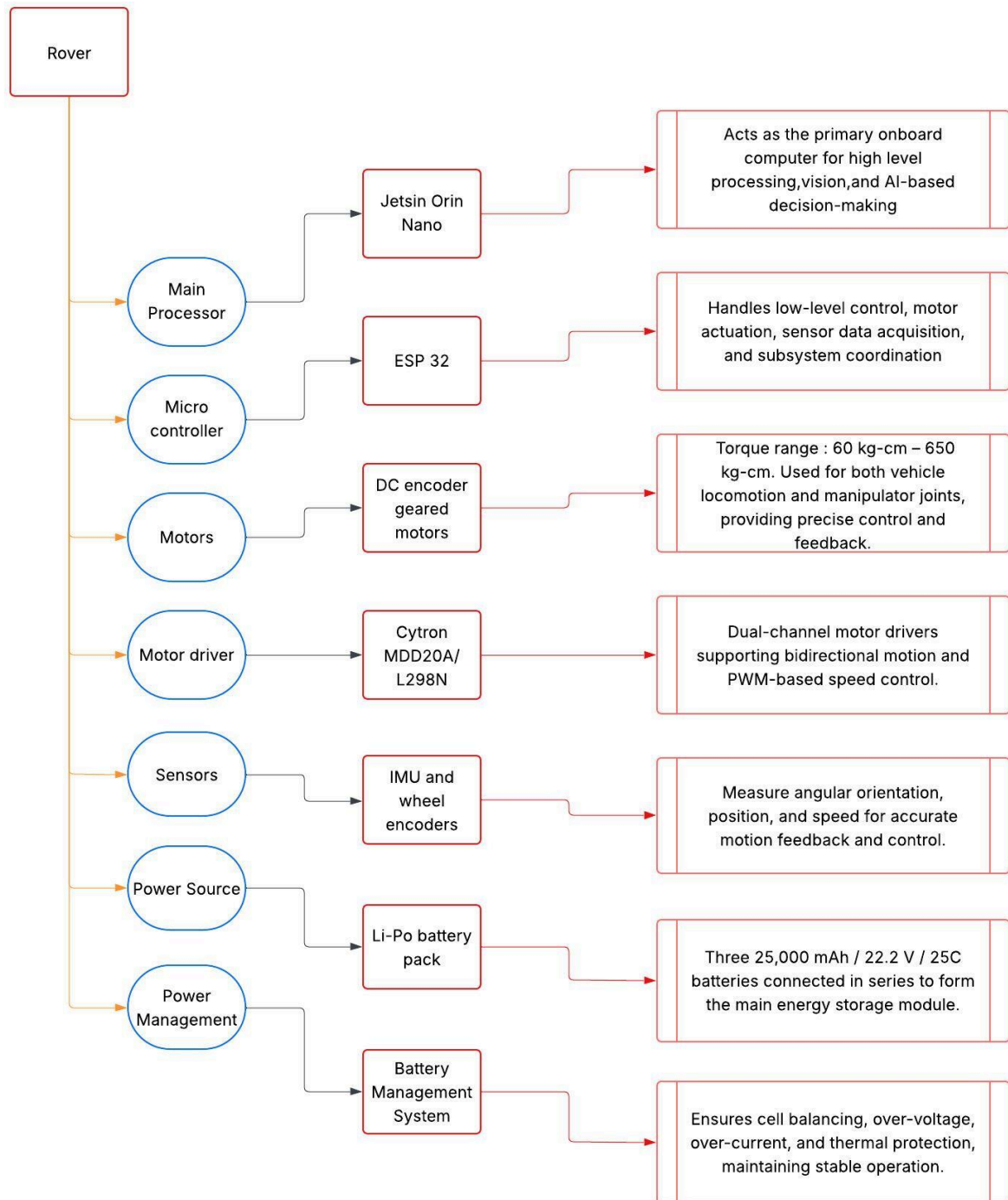


Figure 1.2

5.3 Controls and Locomotion

The locomotion system uses high-torque DC encoder geared motors, with quadrature encoders providing precise speed and position feedback. Motor control is managed through Cytron MDD20A drivers, while an ESP32 with its dual-core LX6 processors handles PWM output and real-time encoder reading. This data is relayed to the Jetson Orin Nano for higher-level navigation. IMU inputs improve motion stability, and a 22.2 V Li-Po battery pack with a BMS ensures safe, consistent power delivery. The setup provides accurate and responsive rover movement.

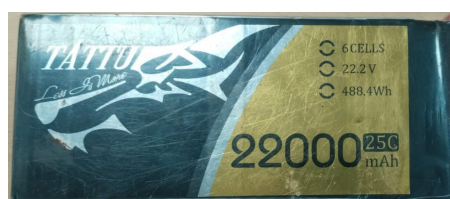


5.4 Manipulator System

The manipulator subsystem employs high-torque, planetary-geared actuators with closed-loop control using high-resolution quadrature encoders for precise joint state estimation. End-effector vision, enabled through an onboard camera, supports visual servoing, pose estimation, and grasp planning, while arm motion is computed through inverse kinematics (IK) and executed via trajectory-controlled joint commands. This architecture enables accurate, stable, and perception-guided manipulation.

5.5 Power Management

The power system uses a 22.2 V, 25C, 22,000 mAh Li-Po pack with a built-in multi-cell BMS that supports ~4 hours of operation. The BMS integrates protection and sensing ICs for cell balancing, over-voltage, over-current, and thermal management, while buck-boost converters generate regulated rails for compute, motor drivers, and sensors.



5.6 Bio Assembly System

The Astrobiology Module's electronics are centered on three motors for its primary functions: one for drilling to collect samples, one for regulating chemical flow during analysis, and one for the movement of the entire module. A motor driver provides the necessary power to run these motors reliably. The entire operation is governed by an ESP microcontroller, which manages the control signals, processes data, and handles communication with the main rover systems.

5.7 Communication and Navigation

The rover's Communications & Navigation system uses an Antenna for all data and command transmission, enabling communication with Earth or an orbiter. Multiple webcams capture general visual data for teleoperation and documentation, providing a broad view of the terrain. Crucially, the ZED camera is a specialized stereo camera that provides 3D depth perception and visual odometry, which is vital for autonomous navigation, obstacle avoidance, and creating real-time 3D maps of the surrounding environment.

5.8 PCB Design

A Printed Circuit Board (PCB) serves as the physical platform and electrical backbone for most electronic devices. It uses etched copper traces to reliably route signals and power between mounted components like ICs and resistors, eliminating manual wiring. This structure provides a compact, mechanically stable, and electrically sound foundation essential for reliable device performance.